F18CE2 MV Calculus and Real Analysis B: Tutorial 3

1 Consider the set $A = \{x > 0 : x^2 > 4\}$. Prove that $\inf(A) = 2$.

2 Let (a_n) denote a convergent sequence. Prove that the limit of (a_n) as $n \to \infty$ is unique.

3 Give an example of a convergent sequence that is not monotone. Give an example of a monotone sequence that is not convergent.

4 The sequence (a_n) is such that $2a_{n+1} \leqslant a_n + a_{n+2}$, for all $n \geqslant 1$. Define a new sequence (b_n) by $b_n = a_{n+1} - a_n$. Show that the sequence (b_n) is increasing.

5 When does an increasing sequence converge? When does a decreasing sequence converge?

6 Let $a_n = 2^n/n!$. By considering the ratio a_{n+1}/a_n, show that (a_n) is decreasing. By finding a suitable bound on (a_n), prove that (a_n) converges.

7 Let $a_n = n^3/8^n$. By considering the ratio a_{n+1}/a_n, show that (a_n) is decreasing. By finding a suitable bound on (a_n), prove that (a_n) converges.

8 Let (x_n) be an increasing sequence with $x_1 \geqslant 0$. Define a new sequence (a_n) by

$$a_n = \frac{2x_n + 3}{x_n + 2}$$

By considering $a_{n+1} - a_n$, show that (a_n) is increasing.

By considering $2 - a_n$, find a bound on (a_n) and hence show that (a_n) is convergent.